

# Concept Map: Definitions and Results

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## Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `>` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
library(broom) # Summarizes key information about statistical objects in tidy tibbles
library(broom.helpers) # Provides suite of functions to work with regression model 'broom::tidy()' t
```

Here are some R settings I use in this document:

```
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
```

```

ggplot2::theme_bw() +
  # ggplot2::labs(col = "") +
  ggplot2::theme(
    legend.position = "bottom",
    text = ggplot2::element_text(size = 12, family = "serif"))

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE

```

This appendix shows how the definitions and results (**def**, **thm**, **lem**, **cor**, **prp** callouts) in the notes depend on one another.

We say result *B* is a **descendant** of result *A* if *A* is referenced inside the statement or proof of *B* (directly, or transitively through a chain of intermediate results). The more descendants a result has, the more of the rest of the notes rests on it — so descendant count is a rough measure of how foundational a result is.

The dependency graph is built by `data-raw/callout-graph.R`, which scans the source `.qmd` files and saves the result to `inst/extdata/callout-graph.rds`. This appendix reads that saved file rather than re-scanning the notes on every render, so **re-run that script whenever divs are added, removed, or re-titled** to refresh the diagram and Table 1.

```

library(igraph)
library(ggraph)
library(ggrepel)

cg <- readRDS(here::here("inst/extdata/callout-graph.rds"))

type_levels <- c("def", "thm", "lem", "cor", "prp")
type_labels <- c(
  def = "Definition", thm = "Theorem", lem = "Lemma",
  cor = "Corollary", prp = "Proposition"
)
type_palette <- c(
  def = "#1b9e77", thm = "#d95f02", lem = "#7570b3",
  cor = "#e7298a", prp = "#66a61e"
)

```

There are 472 labeled definitions and results in the notes, connected by 143 direct dependency links.

## 1 Dependency diagram

The dependency diagram is only available in the HTML version of the notes<sup>1</sup>.

## 2 Descendants of each result

Table 1 lists every result that has at least one descendant, sorted by the number of total descendants (direct or transitive).

<sup>1</sup><https://morrison-lab.github.io/rme/chapters/concept-map.html>

```

ranked <- cg$nodes[cg$nodes$n_desc >= 1, ]
ranked <- ranked[order(-ranked$n_desc, ranked$id), ]

descendant_table <- data.frame(
  Result      = ranked$title,
  Type        = type_labels[ranked$type],
  `Direct descendants` = ranked$n_direct,
  `Total descendants`  = ranked$n_desc,
  check.names = FALSE, row.names = NULL, stringsAsFactors = FALSE
)

knitr::kable(descendant_table, format = "pipe", align = "llrr")

```

Table 1: All results with at least one descendant, sorted by number of total descendants (most-foundational first).

| Result                                                        | Type            | Direct<br>descendants | Total<br>descendants |
|---------------------------------------------------------------|-----------------|-----------------------|----------------------|
| Probability measure                                           | Defini-<br>tion | 7                     | 39                   |
| Odds                                                          | Defini-<br>tion | 4                     | 20                   |
| Odds as a function of probability                             | Theorem         | 5                     | 19                   |
| log-odds                                                      | Defini-<br>tion | 4                     | 17                   |
| Expectation, expected value, population mean                  | Defini-<br>tion | 5                     | 8                    |
| Odds from log-odds                                            | Lemma           | 2                     | 7                    |
| Probability as a function of odds                             | Theorem         | 4                     | 7                    |
| Conditional probability                                       | Defini-<br>tion | 4                     | 6                    |
| Derivative of odds function                                   | Theorem         | 3                     | 6                    |
| Risk-set score total                                          | Defini-<br>tion | 2                     | 5                    |
| Derivative of log-odds by odds                                | Lemma           | 1                     | 5                    |
| One plus odds in terms of non-event probability               | Corol-<br>lary  | 1                     | 4                    |
| Hazard function, hazard rate, hazard rate function            | Defini-<br>tion | 3                     | 4                    |
| Interval failure probability                                  | Defini-<br>tion | 3                     | 4                    |
| logit function                                                | Defini-<br>tion | 2                     | 4                    |
| Variance                                                      | Defini-<br>tion | 3                     | 4                    |
| Derivative of odds w.r.t. log-odds                            | Lemma           | 2                     | 4                    |
| Derivative of log-odds by odds                                | Theorem         | 1                     | 4                    |
| Proportional-hazards decomposition of the hazard              | Theorem         | 1                     | 4                    |
| Derivative of odds function in terms of odds                  | Corol-<br>lary  | 1                     | 3                    |
| Cumulative distribution function (CDF)                        | Defini-<br>tion | 2                     | 3                    |
| Statistical independence                                      | Defini-<br>tion | 3                     | 3                    |
| Proportional-hazards first-order interval failure probability | Defini-<br>tion | 1                     | 3                    |

Table 1: All results with at least one descendant, sorted by number of total descendants (most-foundational first).

| Result                                                     | Type            | Direct<br>descendants | Total<br>descendants |
|------------------------------------------------------------|-----------------|-----------------------|----------------------|
| Cox proportional hazards model                             | Defini-<br>tion | 1                     | 3                    |
| Vector derivative                                          | Defini-<br>tion | 1                     | 3                    |
| Point-mass optimality of the profile baseline hazard       | Lemma           | 1                     | 3                    |
| Derivative of log-odds by probability                      | Theorem         | 2                     | 3                    |
| Law of conditional probability                             | Theorem         | 2                     | 3                    |
| An event and its complement sum to 1                       | Theorem         | 1                     | 3                    |
| Complement rule                                            | Corol-<br>lary  | 2                     | 2                    |
| Covariance                                                 | Defini-<br>tion | 2                     | 2                    |
| Cox-Snell generalized residuals                            | Defini-<br>tion | 2                     | 2                    |
| Dot product/linear combination/inner product               | Defini-<br>tion | 2                     | 2                    |
| Riemann integrable                                         | Defini-<br>tion | 2                     | 2                    |
| Cox PH partial likelihood                                  | Defini-<br>tion | 2                     | 2                    |
| Risk Score                                                 | Defini-<br>tion | 1                     | 2                    |
| Derivative of log-odds with respect to coefficients        | Lemma           | 2                     | 2                    |
| Simplified expression for inverse odds function            | Lemma           | 1                     | 2                    |
| One minus inverse-odds                                     | Lemma           | 2                     | 2                    |
| Profile likelihood in terms of point masses                | Lemma           | 2                     | 2                    |
| Single-subject failure probability in a risk set           | Lemma           | 2                     | 2                    |
| Derivative of odds in terms of probability                 | Theorem         | 1                     | 2                    |
| Derivative of inverse odds function                        | Theorem         | 1                     | 2                    |
| Derivative of a linear map                                 | Theorem         | 1                     | 2                    |
| Expanded expression for logit                              | Theorem         | 1                     | 2                    |
| Log-odds as a function of probability                      | Theorem         | 1                     | 2                    |
| Log-odds via the logit function                            | Corol-<br>lary  | 1                     | 1                    |
| Odds via the odds function                                 | Corol-<br>lary  | 1                     | 1                    |
| Akaike Information Criterion (AIC)                         | Defini-<br>tion | 1                     | 1                    |
| Antiderivative                                             | Defini-<br>tion | 1                     | 1                    |
| Conditional expectation                                    | Defini-<br>tion | 1                     | 1                    |
| conditional hazard                                         | Defini-<br>tion | 1                     | 1                    |
| Continuous function                                        | Defini-<br>tion | 1                     | 1                    |
| Variance/covariance of a $p \times 1$ random vector        | Defini-<br>tion | 1                     | 1                    |
| Deviation from a population or subpopulation<br>mean       | Defini-<br>tion | 1                     | 1                    |
| Deviation of an observation from its subpopulation<br>mean | Defini-<br>tion | 1                     | 1                    |

Table 1: All results with at least one descendant, sorted by number of total descendants (most-foundational first).

| Result                                                      | Type            | Direct<br>descendants | Total<br>descendants |
|-------------------------------------------------------------|-----------------|-----------------------|----------------------|
| Difference in covariate patterns                            | Defini-<br>tion | 1                     | 1                    |
| Difference in log-odds                                      | Defini-<br>tion | 1                     | 1                    |
| Estimated Risk Score                                        | Defini-<br>tion | 1                     | 1                    |
| Expectation of a random matrix                              | Defini-<br>tion | 1                     | 1                    |
| First-order interval failure probability                    | Defini-<br>tion | 1                     | 1                    |
| Hat matrix                                                  | Defini-<br>tion | 1                     | 1                    |
| Per-interval likelihood factor                              | Defini-<br>tion | 1                     | 1                    |
| inverse odds function                                       | Defini-<br>tion | 1                     | 1                    |
| Martingale Residual                                         | Defini-<br>tion | 1                     | 1                    |
| Predicted value                                             | Defini-<br>tion | 1                     | 1                    |
| Quadratic form                                              | Defini-<br>tion | 1                     | 1                    |
| Risk Set                                                    | Defini-<br>tion | 1                     | 1                    |
| Survival function                                           | Defini-<br>tion | 1                     | 1                    |
| Censored-data likelihood                                    | Lemma           | 1                     | 1                    |
| The covariance of a variable with itself is its<br>variance | Lemma           | 1                     | 1                    |
| Difference in log-odds                                      | Lemma           | 1                     | 1                    |
| Difference of log-hazards between two covariate<br>patterns | Lemma           | 1                     | 1                    |
| General formula for odds ratios in logistic<br>regression   | Lemma           | 1                     | 1                    |
| Profile maximum likelihood estimate of each point<br>mass   | Lemma           | 1                     | 1                    |
| The whether-a-failure-occurs factor                         | Lemma           | 1                     | 1                    |
| Exact discretization of the survival factor                 | Lemma           | 1                     | 1                    |
| Chain rule                                                  | Theorem         | 1                     | 1                    |
| Derivative of probability w.r.t. log-odds                   | Theorem         | 1                     | 1                    |
| Derivative of a dot product                                 | Theorem         | 1                     | 1                    |
| Derivative of natural logarithm                             | Theorem         | 1                     | 1                    |
| Vector-derivative of a matrix-vector product                | Theorem         | 1                     | 1                    |
| Expressions for expit function                              | Theorem         | 1                     | 1                    |
| Fubini's theorem (Riemann version)                          | Theorem         | 1                     | 1                    |
| Mean Squared Error equals Bias Squared plus<br>Variance     | Theorem         | 1                     | 1                    |
| Odds ratios are reversible                                  | Theorem         | 1                     | 1                    |
| Probability as a function of log-odds                       | Theorem         | 1                     | 1                    |
| Mean and variance of residuals                              | Theorem         | 1                     | 1                    |
| Law of total probability                                    | Theorem         | 1                     | 1                    |
| Variance of a linear combination                            | Theorem         | 1                     | 1                    |